Legendre ODEs and Legendre Polynomials

1 Infinite Series Solutions of the Legendre ODE

We study series solutions of

$$(1-t^2)\,y''(t) - 2t\,y'(t) + \ell(\ell+1)\,y(t) = 0. \tag{1}$$

Two natural centers are an ordinary point ($t=0$) and the regular singular point ($t=1$).

1.1 Power series about $t=0$ (ordinary point)

Seek

$$y(t) = \sum_{k=0}^{\infty} a_k t^k, \qquad y'(t) = \sum_{k=1}^{\infty} k a_k t^{k-1}, \qquad y''(t) = \sum_{k=2}^{\infty} k(k-1) a_k t^{k-2}.$$

Plugging into (1) and matching powers gives, for all $k \geq 0$,

$$(k+2)(k+1)\,a_{k+2} + \big(\ell(\ell+1) - k(k+1)\big)\,a_k = 0, \tag{2}$$

i.e.

$$a_{k+2} = \frac{k(k+1) - \ell(\ell+1)}{(k+1)(k+2)}\,a_k. \tag{3}$$

Thus even and odd coefficients decouple:

$$\begin{aligned} a_{2(m+1)} &= \frac{(2m)(2m+1) - \ell(\ell+1)}{(2m+1)(2m+2)}\,a_{2m}, \\ a_{2(m+1)+1} &= \frac{(2m+1)(2m+2) - \ell(\ell+1)}{(2m+2)(2m+3)}\,a_{2m+1}. \end{aligned}$$

There are two linearly independent analytic solutions on $|t|<1$, one even (from a_0) and one odd (from a_1). The radius of convergence is 1, the distance to the nearest singularities at $t=\pm 1$.

Polynomial truncation. If $\ell = n \in \mathbb{N}$, the numerator in (3) vanishes at $k=n$, so $a_{n+1}=0$ and all higher coefficients are 0. This yields the Legendre polynomial P_n (parity matches n).

Closed products. For $m \geq 1$,

$$a_{2m} = a_0 \prod_{j=0}^{m-1} \frac{(2j)(2j+1) - \ell(\ell+1)}{(2j+1)(2j+2)}, \qquad a_{2m+1} = a_1 \prod_{j=0}^{m-1} \frac{(2j+1)(2j+2) - \ell(\ell+1)}{(2j+2)(2j+3)}.$$

When $\ell = n \in \mathbb{N}$, these products terminate.

1.2 Frobenius expansion about $t=1$ (regular singular point)

Let $z = 1-t$ so that $t = 1-z$. Then

$$(1-t^2)y'' - 2ty' + \ell(\ell+1)y = 0 \quad \Longleftrightarrow \quad z(2-z)\,y_{zz} + 2(1-z)\,y_z + \ell(\ell+1)\,y = 0,$$

where subscripts denote z–derivatives.

Indicial equation. In standard form $y'' + p(t)y' + q(t)y = 0$ with $p(t) = -\frac{2t}{1-t^2}$, $q(t) = \frac{\ell(\ell+1)}{1-t^2}$, at $t = 1$ we have

$$\alpha = \lim_{t\to 1}(t-1)p(t) = 1, \qquad \beta = \lim_{t\to 1}(t-1)^2 q(t) = 0,$$

so $r(r-1) + \alpha r + \beta = 0$ gives a double root $r = 0$. Hence one analytic solution and a logarithmic companion.

Analytic solution at $t = 1$. Seek $y_1(z) = \sum_{k=0}^{\infty} c_k z^k$ with $c_0 \neq 0$. Substituting yields, for $k \geq 0$,

$$2(k+1)^2\, c_{k+1} + \big(\ell(\ell+1) - k(k+1)\big)\, c_k = 0, \quad \text{i.e.} \quad c_{k+1} = \frac{k(k+1) - \ell(\ell+1)}{2(k+1)^2}\, c_k. \tag{4}$$

With $c_0 = 1$, $y_1(1) = 1$. If $\ell = n \in \mathbb{N}$, then $c_{n+1} = 0$ and the series truncates to $P_n(t)$.

Logarithmic companion near $t = 1$. Because the indicial root is repeated, a second solution has the Frobenius form

$$y_2(z) = y_1(z)\, \log z + \sum_{k=0}^{\infty} b_k z^k.$$

Using the differential operator $\mathcal{L}[y] = z(2-z)y_{zz} + 2(1-z)y_z + \ell(\ell+1)y$ and that $\mathcal{L}[y_1] = 0$, a short computation gives

$$\mathcal{L}\Big[\sum_{k\geq 0} b_k z^k\Big] = 2(2-z)\, y_1'(z) - y_1(z).$$

Equating coefficients (with $y_1(z) = \sum c_k z^k$) yields, for all $k \geq 0$,

$$2(k+1)^2\, b_{k+1} + \big(\ell(\ell+1) - k(k+1)\big)\, b_k \;=\; 4(k+1)c_{k+1} - (2k+1)c_k, \tag{5}$$

with a convenient normalization $b_0 = 0$ to ensure linear independence. This y_2 represents the (local) Legendre function of the second kind Q_ℓ, exhibiting the logarithmic singularity at $t = 1$.

Convergence. The z–series for y_1 has radius up to the next singularity at $t = -1$, i.e. $|z| < 2$; the analytic part of y_2 has the same radius (the $\log z$ factor reflects the repeated indicial root).

Remark 1.1 (Summary)**.**

- *About $t = 0$: two analytic series (even/odd) with recurrence* (3)*, convergent for $|t| < 1$; truncation to P_ℓ when $\ell \in \mathbb{N}$.*
- *About $t = 1$: one analytic series with recurrence* (4) *and a logarithmic companion governed by* (5)*; for integer ℓ, the analytic series truncates to P_ℓ.*

2 Derivation of Legendre polynomials

2.1 The Starting Identity

Let

$$W(t) = (t^2 - 1)^n.$$

It is straightforward to verify that W satisfies the first-order relation

$$(t^2 - 1)W'(t) - 2ntW(t) = 0. \tag{6}$$

This is our point of departure for constructing Legendre polynomials.

2.2 Differentiating the Relation

We differentiate (6) repeatedly. Using Leibniz' rule, for any integer $k \geq 0$ we obtain

$$\frac{d^k}{dt^k}\big[(t^2-1)W'(t) = (t^2-1)W^{(k+1)}(t) + 2k\,t\,W^{(k)}(t) + k(k-1)W^{(k-1)}(t),$$
$$\frac{d^k}{dt^k}\big[tW(t) = tW^{(k)}(t) + kW^{(k-1)}(t).$$

Hence, after differentiating (6) k times we arrive at

$$(t^2-1)W^{(k+1)} + 2k\,t\,W^{(k)} + k(k-1)W^{(k-1)} - 2n\big(tW^{(k)} + kW^{(k-1)}\big) = 0. \tag{7}$$

2.3 Specializing to $k = n$

Substituting $k = n$ into (7), the terms involving $tW^{(n)}$ cancel, leaving

$$(t^2-1)W^{(n+1)} - n(n+1)W^{(n-1)} = 0. \tag{8}$$

This is a crucial simplification.

2.4 Differentiating Once More

Differentiating (8) with respect to t yields

$$(2t)W^{(n+1)} + (t^2-1)W^{(n+2)} - n(n+1)W^{(n)} = 0,$$
$$(t^2-1)W^{(n+2)} + 2tW^{(n+1)} - n(n+1)W^{(n)} = 0. \tag{8$'$}$$

2.5 Legendre Polynomials

We now define the Legendre polynomial P_n via Rodrigues' formula:

$$P_n(t) := \frac{1}{2^n n!}W^{(n)}(t) = \frac{1}{2^n n!}\frac{d^n}{dt^n}\big[(t^2-1)^n\big].$$

Thus,

$$W^{(n)} = 2^n n!\,P_n(t), \quad W^{(n+1)} = 2^n n!\,P_n'(t), \quad W^{(n+2)} = 2^n n!\,P_n''(t).$$

Substituting these into the previous identity gives

$$(t^2-1)P_n''(t) + 2tP_n'(t) - n(n+1)P_n(t) = 0.$$

Multiplying through by -1 we obtain the **Legendre differential equation**:

$$(1-t^2)P_n''(t) - 2tP_n'(t) + n(n+1)P_n(t) = 0. \tag{9}$$

2.6 Remarks

- The above derivation shows precisely how repeated differentiation of the basic identity (6) yields the Legendre ODE.
- The Legendre polynomials P_n are therefore *special solutions* of (9), which is a Sturm–Liouville eigenvalue problem.
- Their orthogonality on $[-1, 1]$, recurrence relations, and generating function can all be deduced from this foundation.

3 Mathematical significance of Legendre polynomials

The Legendre differential equation

$$(1-t^2)y''(t) - 2ty'(t) + n(n+1)y(t) = 0 \tag{10}$$

is a central second-order linear ODE in mathematical physics. Its solutions, the *Legendre polynomials*, form one of the most important families of orthogonal polynomials. They arise naturally in problems with spherical symmetry and have a broad range of applications in analysis, geometry, and applied sciences.

3.1 Orthogonal Polynomials

Legendre polynomials $P_n(t)$ are orthogonal on $[-1,1]$ with respect to the inner product

$$\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt.$$

They thus form a basis for $L^2([-1,1])$, allowing functions to be expanded as series in P_n. This is analogous to Fourier series on the circle.

3.2 Sturm–Liouville Theory

The Legendre equation is an example of a Sturm–Liouville problem, with eigenvalues $n(n+1)$. This framework guarantees real eigenvalues, orthogonality, and completeness, making Legendre polynomials fundamental in spectral theory.

3.3 Laplace's Equation in Spherical Coordinates

When Laplace's equation $\Delta u = 0$ is solved in spherical coordinates, separation of variables leads to the Legendre ODE. Its solutions, the Legendre polynomials, describe the angular dependence of harmonic functions with spherical symmetry.

3.4 Electrostatics and Gravitational Potentials

Multipole expansions in electrostatics and Newtonian gravity involve Legendre polynomials. The potential due to a charge or mass distribution can be expressed as a series in $P_n(\cos\theta)$, reflecting angular dependence around a source.

3.5 Quantum Mechanics

In the Schrödinger equation for the hydrogen atom, the angular part reduces to spherical harmonics $Y_n^m(\theta,\phi)$. These are built from associated Legendre functions, generalizations of P_n, making Legendre theory indispensable in atomic physics.

3.6 Other Applications

Legendre functions also appear in:

- Geophysics: modeling gravitational and magnetic fields of Earth.
- Astrophysics: describing gravitational potentials of planets and stars.
- Numerical analysis: Gaussian quadrature rules use roots of P_n for optimal integration accuracy.